Bio-Inspired Locomotion for *Walter* (Walking Alternating Tripod for Evolutionary Research): CAD, Firmware, Optimization, and MuJoCo PPO

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Abstract

Walter (**W**alking **A**lternating **T**ripod for **E**volutionary **R**esearch) is a twelve-degree-of-freedom hexapod built for this Engineering 90 capstone. The team fabricated a **complete hardware platform**: CAD-designed, 3D-printed body, legs, and linkages; hobby servos; Teensy 4.1 and Pololu eighteen-channel driver; SparkFun BMI270 IMU; power regulation; and three iterations of custom *Hexapod V2* PCBs. On the control side, the project **completed two development tracks**: (i) open-loop gait optimization in a reduced-order model with deterministic export to firmware (`hexapod_openloop_sine`), and (ii) a MuJoCo [1] PPO [2] training and evaluation pipeline (Stable-Baselines3 [3]) with stored checkpoints, configs, and short evaluation summaries for representative locomotion experiments. Sec. 9 summarizes surrogate metrics, exported gait parameters, a MuJoCo model view, and sample PPO evaluation means. The team **did not** implement the full layered adaptive firmware described in internal phase requirements (IMU-closed-loop reflexes and online adaptation on the MCU); that stack remains **design groundwork** for future work. Overall, the capstone establishes a **reproducible hexapod platform** and defensible optimization-to-firmware and simulation-based RL workflows; materials are at `https://github.com/HowardWHSrun/WALTER`. **Purchased cost:** electronics (three BOMs) US$595.20; three PCB orders US$197.47; **grand total US$792.67** (one PCB order awaited payment at writing). Filament and fasteners are outside this sum.

1 Introduction

Reliable legged locomotion requires coordinating periodic limb motion with feedback under limited sensing and computation. Insects organize locomotor control in *layers* that are standard in the literature: rhythmic coordination (often modeled as CPG-like structure), fast sensory feedback, and slower parameter adjustment [4, 5]. This capstone uses that organization as **bio-inspired control motivation**, not as a claim that every layer is already running on hardware.

Framing of contributions. The strongest completed work is **platform and methods**: a reproducible hexapod with CAD and custom PCBs; an **open-loop** gait optimization loop that exports parameters to firmware; and a **MuJoCo PPO** training and evaluation workflow with stored checkpoints and configs. A **layered adaptive** architecture (IMU fusion, reflexes, slow adaptation) is **specified in design documents** and informs observation choices in simulation, but it is **not fully implemented** on the MCU. The report therefore emphasizes what was built and verified, and treats full adaptive locomotion on the robot as **future engineering**, not an accomplishment of this submission.

Bio-inspired elements (precise). The **alternating tripod** is a biologically motivated low-dimensional gait prior: two stance groups alternate before any learning is applied. **IMU-based body-state cues** (roll, pitch, rates) match how many legged systems use gravito-inertial information for posture [6, 7] and align the simulation observation stack with how *Walter* is instrumented, which matters for eventual sim-to-real. **Reinforcement learning in MuJoCo** explores dynamics and reward shaping under that prior; it does not, by itself, close the hardware adaptive loop. **Surrogate optimization** of a compact sinusoidal law isolates an interpretable rhythm before sensor-heavy layers are added—consistent with separating pattern generation from feedback in many biological models [5], but realized here only through open-loop firmware.

Platform. The physical system is *Walter*, a twelve-DOF hexapod with 3D-printed structure, servos, Teensy, driver, IMU, and *Hexapod V2* PCBs (Sec. 6.5–6.6).

Code and report organization. Source code and this document are maintained at `https://github.com/HowardWHSrun/WALTER`. Sec. 2 states the optimization objective and the PPO machinery used in software. Sec. 3–5 and Table 1 summarize three tracks: **(1)** surrogate optimization to firmware (**completed**); **(2)** PPO locomotion in simulation, with IMU-capable observation modes where configured (**completed** as a training/evaluation stack; limited hardware deployment); **(3)** vision-augmented distance objectives (**future work**).

2 Mathematical background: optimization and reinforcement learning

2.1 Reduced-order gait optimization (Track 1)

2.1.1 Decision variables and bounds

The optimizer acts on a vector $\boldsymbol{\phi} \in \mathbb{R}^{25}$ matching `hexapod_no_imu_optimization.py`. Indices are concatenated as

$$\boldsymbol{\phi} = \left(\boldsymbol{\theta}_{\mathrm{abd,rest}}^{\top}, \boldsymbol{\theta}_{\mathrm{flx,rest}}^{\top}, \boldsymbol{\delta}_{\mathrm{abd}}^{\top}, \boldsymbol{\delta}_{\mathrm{flx}}^{\top}, f\right)^{\top}, \tag{1}$$

where each of $\boldsymbol{\theta}_{\mathrm{abd,rest}}, \boldsymbol{\theta}_{\mathrm{flx,rest}}, \boldsymbol{\delta}_{\mathrm{abd}}, \boldsymbol{\delta}_{\mathrm{flx}} \in \mathbb{R}^6$ are in *degrees* in the code (converted to radians internally), indexed by leg order FL, FR, ML, MR, RL, RR, and $f \in \mathbb{R}_+$ is the common gait frequency in hertz. Box constraints are implemented as `scipy.optimize.Bounds`: for example $f \in [0.4,\ 4.0]\,\mathrm{Hz}$; resting abduction per joint in $[-25°, 25°]$; resting flexion in $[10°, 65°]$; phase shifts in $[-120°, 120°]$. These define a feasible orthotope $\boldsymbol{\phi}_{\min} \leq \boldsymbol{\phi} \leq \boldsymbol{\phi}_{\max}$.

2.1.2 Open-loop joint commands (sinusoidal law)

Let $\omega = 2\pi f$. For leg $\ell \in \{1, \ldots, 6\}$ with fixed tripod offset $\psi_\ell \in \{0, \pi\}$ (alternating tripod), time $t \geq 0$, and fixed amplitude tables $\bar{a}_{\mathrm{abd},\ell}$, $\bar{a}_{\mathrm{flx},\ell}$ (defaults in code, not decision variables), the commanded abduction and flexion angles are

$$\phi_{\mathrm{abd},\ell}(t) = \theta_{\mathrm{abd,rest},\ell} + \bar{a}_{\mathrm{abd},\ell} \sin\!\big(\omega t + \psi_\ell + \delta_{\mathrm{abd},\ell}\big), \tag{2}$$

$$\phi_{\mathrm{flx},\ell}(t) = \theta_{\mathrm{flx,rest},\ell} + \bar{a}_{\mathrm{flx},\ell} \sin\!\big(\omega t + \psi_\ell + \delta_{\mathrm{flx},\ell}\big) - \Delta_{\mathrm{lift},\ell}(t), \tag{3}$$

where $\Delta_{\mathrm{lift},\ell}$ implements swing-phase foot clearance (subtracting flexion when the abduction sine is positive), identical in structure to the firmware `LIFT_SWING_DEG_PEAK` term. Velocities $\dot{\phi}_{\mathrm{abd},\ell}$, $\dot{\phi}_{\mathrm{flx},\ell}$ are obtained by differentiation and feed the planar surrogate dynamics.

2.1.3 Surrogate planar dynamics and simulation

The reduced model integrates an eight-dimensional state

$$\mathbf{x}(t) = \big(x,\ \dot{x},\ z,\ \dot{z},\ \phi_{\mathrm{roll}},\ \dot{\phi}_{\mathrm{roll}},\ \phi_{\mathrm{pitch}},\ \dot{\phi}_{\mathrm{pitch}}\big)^\top, \tag{4}$$

with $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\phi})$ assembled from lumped masses, soft ground contact, leg thrust and support forces derived from $\phi_{\mathrm{abd},\ell}$, $\phi_{\mathrm{flx},\ell}$ and contact weights, and linear damping on body motion and attitude rates (`body_ode` in the script). For fixed $\boldsymbol{\phi}$, the trajectory is obtained by `solve_ivp` (Runge–Kutta) over $t \in [0, T]$ with T the simulation horizon (`sim_duration_s`), sampled at N time points (`sample_count`).

2.1.4 Scalar objective (explicit weighted sum)

Let a subscript "ss" denote statistics evaluated on a *steady-state* suffix of the horizon (the code uses the last 60% of samples for oscillation metrics and mean speed). Write $\bar{v}_x$ for mean forward velocity $\mathbb{E}_{\mathrm{ss}}[\dot{x}]$, and RMS values $\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{roll}}]$, $\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{pitch}}]$, $\mathrm{RMS}_{\mathrm{ss}}[z]$. Let $q_{\mathrm{servo}} \geq 0$ be a slack on peak torque proxy exceeding a threshold (0.85 of stall in the implementation), and $\sigma_{\mathrm{contact}}$ the time mean of the per-timestep standard deviation of contact weights across legs. With nonnegative weights w. matching `OptimizationConfig`, the implemented cost is

$$\begin{aligned}\mathcal{J}(\boldsymbol{\phi}) = -w_{\mathrm{fwd}}\,\bar{v}_x + w_{\mathrm{back}}\left[\max(0,-\bar{v}_x)\right]^2 \\ + w_{\mathrm{rp}}\left(\mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{roll}}]^2 + \mathrm{RMS}_{\mathrm{ss}}[\phi_{\mathrm{pitch}}]^2\right) + w_z\,\mathrm{RMS}_{\mathrm{ss}}[z]^2 \\ + w_{\mathrm{servo}}\,q_{\mathrm{servo}}^2 + w_{\mathrm{contact}}\,\sigma_{\mathrm{contact}}^2 + w_{\mathrm{reg}}\,\|\boldsymbol{\phi}-\boldsymbol{\phi}_0\|_{\mathrm{scaled}}^2,\end{aligned} \tag{5}$$

where $\boldsymbol{\phi}_0$ is the default parameter vector and $\|\cdot\|_{\mathrm{scaled}}$ averages coordinatewise squared normalized deviation from the box midpoint (Tikhonov-style regularization keeping $\boldsymbol{\phi}$ in a "reasonable" neighborhood). Optional terms penalize shortfall below a target mean speed if `target_forward_speed_mps` $>$ 0. Infeasible simulations return a large penalty (10^6). **Design intent:** the weighted sum prioritizes forward progress while penalizing backward drift, attitude and vertical oscillation, actuator overload proxies, uneven leg loading, and deviation from nominal $\boldsymbol{\phi}_0$—a deliberate trade that can admit energetic gaits with visible body motion in the surrogate.

2.1.5 Optimization algorithm and restarts

The problem solved in software is

$$\min_{\boldsymbol{\phi}} \; \mathcal{J}(\boldsymbol{\phi}) \quad \text{subject to} \quad \boldsymbol{\phi}_{\min} \le \boldsymbol{\phi} \le \boldsymbol{\phi}_{\max}. \tag{6}$$

L-BFGS-B (`scipy.optimize.minimize`, method `L-BFGS-B`) approximates curvature of $\mathcal{J}$ with a limited-memory inverse Hessian and projects each iterate onto the box. Each evaluation of $\mathcal{J}$ requires one numerical integration of the surrogate. The outer loop runs R random restarts (`restarts`): one start at $\boldsymbol{\phi}_0$ and $R-1$ uniform samples in the box; the best $\boldsymbol{\phi}^\star$ by lowest $\mathcal{J}$ is retained. Limits `maxiter`, `maxfun`, and `ftol` cap work per restart.

The **code path** from optimization to firmware is deterministic: $\boldsymbol{\phi}^\star$ is serialized to `no_imu_optimization_summary.` `sync_gait_params_from_json.py` emits `optimized_gait_params.h`; `hexapod_openloop_sine.ino` evaluates the same `leg_commands()` map, applies bench degree-to-pulse gains and clamps, and streams servo targets. Thus optimization-to-code is *algebraic parity*, not a learned policy.

2.2 Policy optimization in MuJoCo (PPO with GAE)

Legged RL is cast as an MDP with MuJoCo supplying transitions; we use actor–critic PPO as implemented in Stable-Baselines3 [3]. **Generalized advantage estimation** (GAE) [2] builds advantages from TD residuals $\delta_t = r_t + \gamma V_\psi(s_{t+1}) - V_\psi(s_t)$:

$$\hat{A}_t = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell\, \delta_{t+\ell}, \tag{7}$$

with λ and discount γ set in YAML. **PPO** [2] stabilizes updates via a clipped ratio between the new and behavior policy. Define the **probability ratio**

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}. \tag{8}$$

For continuous actions, π_θ is typically Gaussian; r_t is the ratio of densities. The **clipped surrogate objective** for a batch of timesteps is

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta), 1-\epsilon, 1+\epsilon\big)\,\hat{A}_t\big)\Big], \tag{9}$$

with clip range $\epsilon \in (0,1)$ (often 0.2). Large policy steps that inflate r_t when $\hat{A}_t > 0$ (or shrink r_t when $\hat{A}_t < 0$) are clipped so the effective objective stops increasing, reducing catastrophic updates common in vanilla policy gradients.

Let $\hat{R}_t$ denote a target return for the critic (e.g., n-step return or GAE-backed value target). With coefficients $c_v, c_e > 0$, the **composite loss** maximized in practice is the negative of

$$\mathcal{L}_{\text{total}}(\theta, \psi) = -\mathcal{L}^{\text{CLIP}}(\theta) + c_v\,\mathbb{E}_t\big[(V_\psi(s_t) - \hat{R}_t)^2\big] - c_e\,\mathbb{E}_t\big[H(\pi_\theta(\cdot \mid s_t))\big], \tag{10}$$

where H is differential entropy for Gaussian policies. Minimizing $\mathcal{L}_{\text{total}}$ corresponds to ascending $\mathcal{L}^{\text{CLIP}}$ while fitting the critic and discouraging collapse of exploratory variance. Updates use minibatches over several epochs on each rollout buffer; hyperparameters (learning rate, rollout length `n_steps`, batch size, γ, λ, ϵ, c_v, c_e) are set in YAML under `hexapod_training_new/configs/`.

3 Track I: Surrogate optimization and firmware export

This track **implements** Sec. 2.1: the $\boldsymbol{\phi}$ parameterization, sinusoidal `leg_commands()`, the eight-state surrogate, cost (5), box-constrained L-BFGS-B with restarts, and a **deterministic path** from $\boldsymbol{\phi}^\star$ to C headers and `hexapod_openloop_sine`. The workflow isolates a tunable open-loop rhythm before IMU-closed-loop layers are required.

What we shipped. Optimizer script, summary JSON, JSON-to-header sync, and open-loop Arduino sketch with exported gains. Quantitative surrogate outputs appear in Tables 6–7; the high-fidelity MuJoCo asset used for RL appears in Fig. 4.

4 Track II: PPO locomotion in MuJoCo

The policy $\pi_\theta(a|s)$ conditions on **proprioception** (joint positions and velocities), simulator **base kinematics**, and—in IMU-focused YAML configs—**accelerometer and gyroscope** channels (or complementary-filter-style stacks) so roll, pitch, and body rates align with how *Walter* is sensed on hardware. **Tripod structure** enters through phase features, structured policy heads (e.g., sCPG variants), or reward shaping, depending on the experiment. Training uses Eqs. (7)–(9) on MuJoCo rollouts.

Scope. The team **built, trained, and evaluated** policies in simulation, archived checkpoints, and wrote short evaluation summaries for representative runs (Table 8). Deploying a trained policy as the primary closed-loop driver on the physical robot—with the same observation latency and actuation path as firmware—was **out of scope** for verification in this report.

5 Track III: Vision-augmented distance objectives (future)

Future work. Augment s with **visual odometry** or pose from an overhead or onboard camera so rewards can emphasize **distance traveled** with less privileged state. Let $p_t \in \mathbb{R}^2$ be horizontal pose from visual integration or fiducials, and consider forward displacement $x_t - x_0$. A natural reward increment is

$$r_t = \alpha\,(x_{t+1} - x_t) - \beta\,\|\omega_t\|^2 - \rho\,\mathrm{penalty}_{\mathrm{fall}} - \kappa\,\mathrm{TV}(a)_t, \tag{11}$$

with IMU rates ω_t penalizing harsh attitude changes, and optional terms for staying upright. The **observation** becomes $s_t = [s_t^{\mathrm{IMU/proprio}}, f_t]$ where f_t is a vector of visual features or estimated pose; PPO then applies the same surrogate (9) with expanded input dimension.

Open challenges include sim-to-real on vision, defining rewards from estimated versus ground-truth motion, and keeping a tripod prior while learning residuals.

Table 1: Summary of the three development tracks for *Walter*.

Track	Technical core	Outcome in this capstone
I	Surrogate $\mathcal{J}(\boldsymbol{\phi})$ (5); L-BFGS-B; JSON→`.h`→ open-loop firmware	**Completed:** optimization exported and flashed path demonstrated
II	PPO (9); GAE (7); MuJoCo env; IMU modes in config	**Completed:** training/eval pipeline and archived runs; hardware policy deployment not fully exercised here
III	Visual or pose features f_t; distance-centric rewards	**Not started:** specification and literature basis only

6 Technical Discussion

6.1 Biological and control context

Decentralized feedback atop rhythmic coordination remains a standard idiom for insect locomotion [4, 5]. Attitude filters for accelerometer–gyro fusion [6, 8] and posture control on terrain [7] inform the IMU observation choices in simulation. Disturbance-triggered transitions [9] motivate reflex layers in the design spec, not a completed hardware demo in this capstone.

6.2 Planned adaptive layer (design specification)

Internal architecture notes (`docs/ARCHITECTURE.md`) describe a **future** two-timescale stack: a **fast tick** ($\sim$10–20 ms) for IMU read, attitude fusion, stability score S, reflex adjustments, gait interpolation, and servo commands; and a **slow tick** (per gait cycle) for windowed costs and optional parameter adaptation. That design informed IMU-inclusive observation modes in simulation; it is **not** claimed as deployed firmware here. The complementary blend

$$\alpha \in (0,1), \quad \theta_{\text{roll}} \leftarrow \alpha(\theta_{\text{roll}} + \omega_x \Delta t) + (1-\alpha)\,\theta_{\text{roll,accel}}, \tag{12}$$

(and analogously for pitch) is the nominal attitude update.

6.3 Open-loop sinusoidal gait (implemented)

For the no-IMU milestone, each leg follows resting abduction and flexion angles plus sinusoidal variation at common frequency f, with tripod group B offset by π from group A. This is the same law optimized in Sec. 2.1 and delivered in Track I (Sec. 3). Amplitudes follow default tables shared by Python and `optimized_gait_params.h`.

6.4 MuJoCo and reinforcement learning (simulation)

High-fidelity training uses Gymnasium + MuJoCo [1] with PPO [2] as implemented in Stable-Baselines3 [3]; the mathematical objective is Eq. (9) with advantages (7). Tracks II–III (Sec. 4–5) specialize the observation and reward; YAML configs set γ, λ, learning rate, network widths, PD gains, and terrain. A representative static render of the simulation model appears in Fig. 4 (Sec. 9.2).

6.5 Mechanical design: CAD and 3D-printed parts

Walter's chassis and legs are built from **parts designed in CAD** and produced by **additive manufacturing** (3D printing): a central body, six leg assemblies, and two-degree-of-freedom linkages per leg (abduction and flexion axes) that mate to the servo output splines. The design prioritizes a repeatable assembly for twelve actuated joints, clearance for wiring, and attachment of the electronics tray and battery. Figure 1 shows an overall render; Figure 2 shows the body, leg, and linkage components used in the assembly. Fasteners and servo hardware complete the mechanical stack; material usage (filament mass) and printed-part cost are not included in the dollar totals in Tables 3–5.

6.6 Electronics, Hexapod V2 PCB, and safety

The bill of materials matches purchased components: Teensy-class MCU (SparkFun 16771), Pololu 1354 eighteen-channel driver, buck converter, BMI270 via Qwiic cabling, dual-cell LiPo and charger, terminals, and servos. Custom *Hexapod V2* PCBs (revision v5 dated 2026-04-03 on the latest order) integrate power and signal routing for Walter's stack; Figure 3 shows a board photo and schematic excerpt. `docs/FIRMWARE.md` records UART to the servo controller, channel ordering FL/FR/ML/MR/RL/RR with abduction then flexion per leg, and pulse mapping $u_k = \mathrm{clamp}(c_k + d_k q_k)$ (typical bounds 600–2400 μs). Open-loop firmware applies degree-to-command gains and clamps before streaming compact-protocol targets; these bench gains are not outputs of the reduced-order optimizer.

7 Requirements and Constraints

7.1 Source of requirements

Engineering requirements are traced to phase exit criteria in `docs/PHASES.md` (Phases 1–4: IMU telemetry, reflexes, cautious mode and recovery, online adaptation), supplemented by safety and interface constraints in `docs/FIRMWARE.md` and the open-loop scope in `firmware/hexapod_openloop_sine/README.m`
`PHASES.md` serves as the team's consolidated requirements-and-constraints specification for firmware behavior.

7.2 Summary: met vs. not met

Satisfied in this project: discrete tripod baseline firmware; open-loop optimization with JSON export and header sync; calibration sketches; CAD-documented mechanical design (Figs. 1–2); a MuJoCo PPO workflow with configs, training scripts, checkpoints, and evaluation writeups. **Not satisfied against the internal phased firmware spec:** Phases 1–4 adaptive behaviors (telemetry-only through online adaptation) are largely **not implemented** on the MCU in the snapshot summarized below.

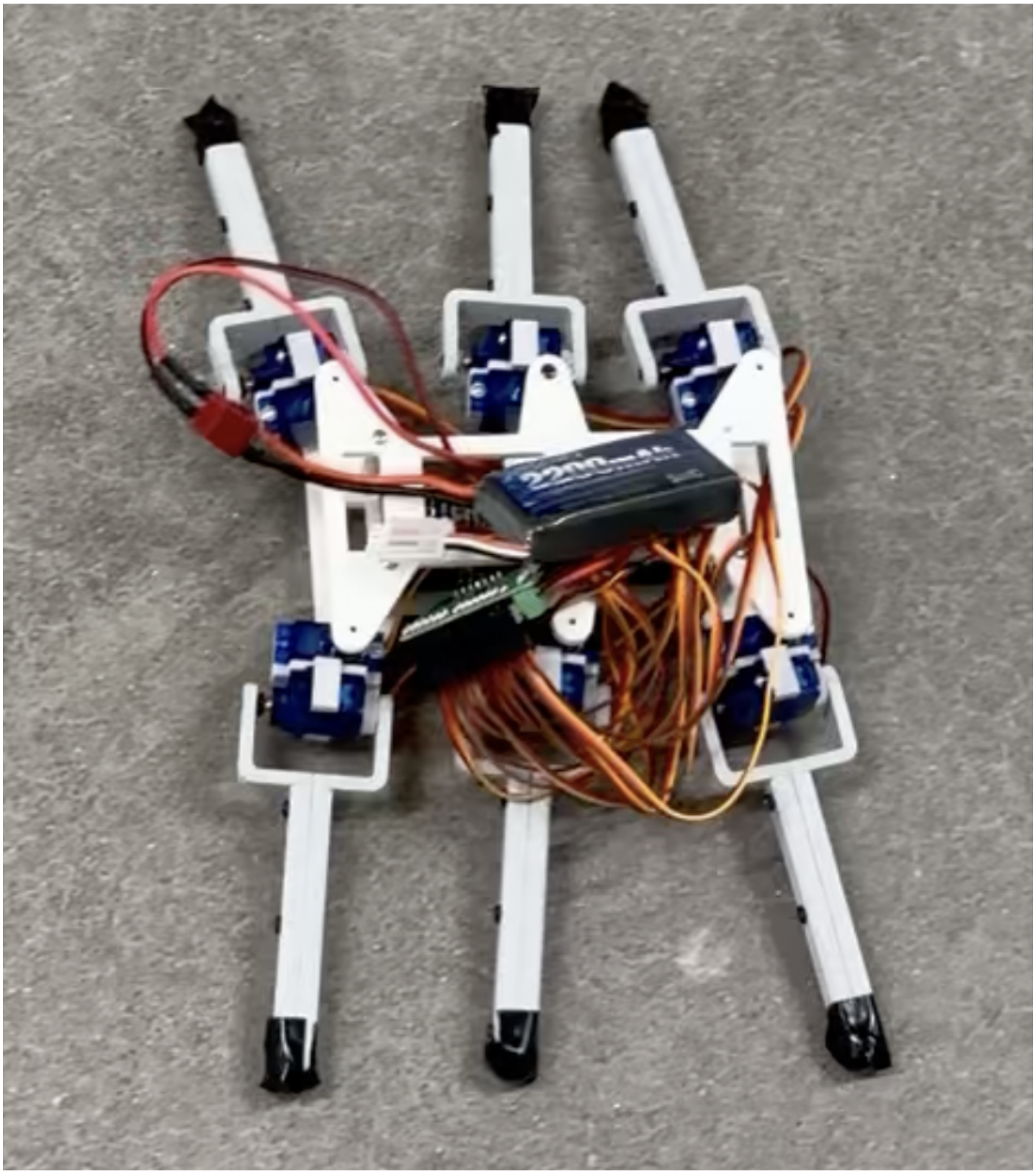
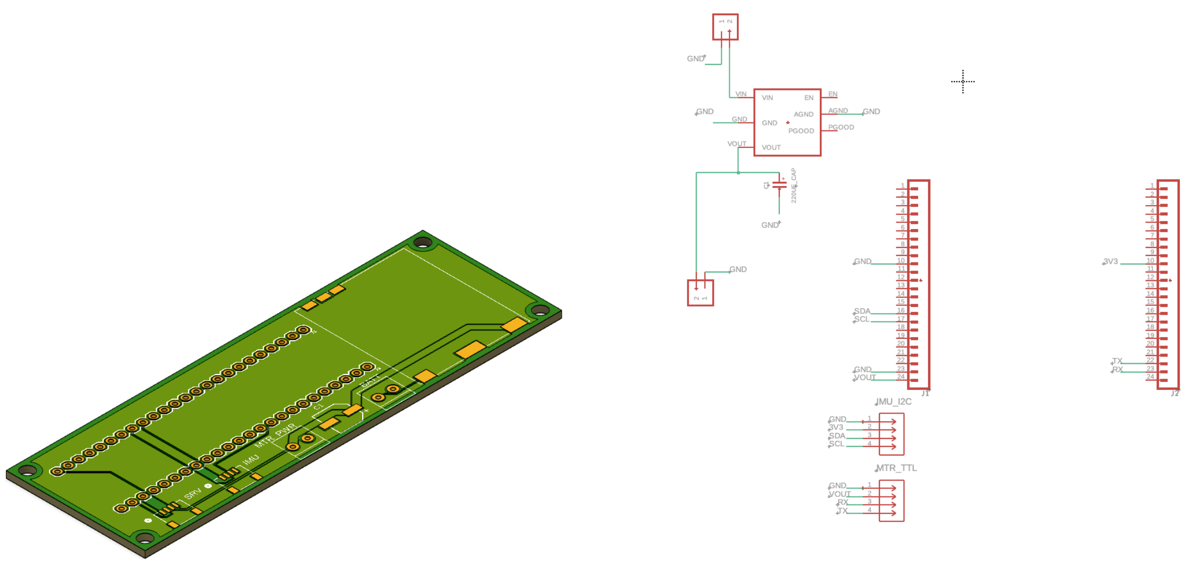


Figure 1: Walter: full-assembly render of the twelve-DOF hexapod (CAD; Walking Alternating Tripod for Evolutionary Research).

(a) Body (b) Leg assemblies (c) Linkage

Figure 2: CAD-designed 3D-printed subassemblies for Walter: chassis, legs, and per-leg linkage.



(a) Assembled PCB (Hexapod V2) (b) Schematic (excerpt)

Figure 3: Custom electronics for Walter: Hexapod V2 board and schematic.

Table 2: Requirement traceability (internal PHASES.md vs. delivered work).

ID	Requirement	Status	Evidence / notes
R1	Phase 1: IMU read, complementary filter, USB telemetry ∼50 Hz; gait unchanged	Not met (in repo)	PHASES.md; folder may be missing
R2	Phase 2: postural reflexes, clamps, serial tuning, EEPROM	Not met (in repo)	ARCHITECTURE.md pseudocode
R3	Phase 3: stability score, cautious mode, push recovery, fall freeze	Not met (in repo)	Documented only
R4	Phase 4: five-parameter adaptation, cost J, SPSA/bandit, EEPROM	Not met (in repo)	Documented only
R5	Hardware: MCU + eighteen-channel driver + servos + documented map + PCBs + Walter mechanical CAD/3D parts	Met	BOM, Figs. 1–3, FIRMWARE.md
R6	Safety: pulse clamps; parameter slew (adaptive layer)	Partial	Clamps in v2/openloop; slew in spec
R7	Open-loop: optimizer/firmware parity; JSON export	Met	Optimizer, sync_gait_params_from_json.py
R8	Simulation: MuJoCo env, PPO training, documented evals	Met	hexapod_training_new/, e90_repo/released_checkpoints/

7.3 Project cost

Tables 3–4 summarize **purchased** electronics (three identical platforms) and **fabrication** costs for three PCB/stencil orders. Table 5 states the **grand total** for those categories only; filament, fasteners, and shop labor are outside the accounting. Servo counts in Table 3 are per platform (eight units listed per BOM line); completing twelve DOF per robot may require additional servos beyond those lines if not already stocked.

8 Methods

8.1 Mechanical assembly (*Walter*)

Printed body and leg components are fastened per the CAD design; each leg receives two servos driving abduction and flexion through the linkage. The custom PCB mounts to the chassis and connects to the Pololu driver and Teensy; power and ground respect the buck converter and screw terminals from the BOM. Bench alignment uses `set_zero` and side-only test sketches before closed-loop or open-loop gait tests.

Table 3: Electronics bill of materials, **one platform** (USD). The team purchased three identical sets.

Part	Description	Vendor	Qty	Unit	Line
SparkFun 16771	Microcontroller	DigiKey	1	37.20	37.20
Pololu 1354	18-channel servo driver	DigiKey	1	46.95	46.95
DFRobot DFR1202	Step-down buck converter	DFRobot	1	12.90	12.90
SparkFun BMI270	6-DoF IMU	DigiKey	1	18.50	18.50
SparkFun 11855	2×7.4 V LiPo + charger	Amazon	1	27.99	27.99
SparkFun 14417	Female Qwiic connector	DigiKey	2	0.60	1.20
SparkFun 17259	100 mm Qwiic cable	DigiKey	1	1.95	1.95
SparkFun 17261	Qwiic cable to female header	DigiKey	1	1.95	1.95
Amphenol YO0201500000G	1×2 screw-down terminal	DigiKey	2	1.28	2.56
DFRobot SER0039	Servo motor	DigiKey	8	5.90	47.20
Subtotal	**one platform**				**198.40**
Electronics total	**three platforms (×3)**				**595.20**

Table 4: PCB and stencil orders, *Hexapod V2* (USD order totals including merchandise, shipping, duties/taxes, and sales tax where charged by the vendor).

Order ID	Description / notes	Status	Total
Y4-11911448A	PCB prototype (5 pcs) + SMT stencil; v5_2026-04-03	Awaiting payment	74.24
W2026030804205410	PCB + stencil shipment	Shipped	68.26
W2026021709554129	PCB + stencil (earlier order)	Shipped	54.97
PCB subtotal			**197.47**

Table 5: Grand total project cost (USD) for reported purchases and fabrication.

Category	Amount
Electronics (three platforms)	595.20
PCB fabrication and stencils (three orders)	197.47
Grand total	**792.67**

8.2 Baseline discrete gait

`firmware/hexapod_walking_v2/hexapod_walking_v2.ino` implements a deterministic tripod state machine with smoothed keyframes (`moveSmoothTo`).

8.3 Reduced-order simulation optimization

`hexapod_brain_roadmap copy/ode_optimization_no_imu/hexapod_no_imu_optimization.py` integrates the surrogate model with `scipy.integrate.solve_ivp`, evaluates the objective on a steady-state window, and calls `scipy.optimize.minimize` (L-BFGS-B). Outputs include `no_imu_optimization_s`
and optional CSV/plots.

8.4 Parameter sync and open-loop firmware

From `firmware/hexapod_openloop_sine/`, run `python3 sync_gait_params_from_json.py` to regenerate `optimized_gait_params.h`. The sketch evaluates the sinusoidal law at `INTERP_DELAY_MS` (default 10 ms), applies startup frequency ramping, maps degrees to integer commands, clamps, and sends Pololu compact-protocol targets on `Serial1`.

8.5 Bench testing and calibration

`firmware/set_zero/` and side-only test sketches support calibration. The open-loop README documents sign and scale tuning when feet drag or the robot shuffles backward.

8.6 MuJoCo PPO training and evaluation

The project produced a **reusable training workflow**: `train.py` reads YAML from `configs/`, writes summaries, checkpoints, and logs under runs/<run_name>/, and pairs with `evaluate.py` using the same configuration. Released material includes zipped policies, per-run evaluation folders with `REPORT.md` summaries, and metadata for reproducibility and reporting.

9 Results

9.1 Reduced-order optimization (surrogate model)

Table 6 summarizes `hexapod_brain_roadmap copy/ode_optimization_no_imu/no_imu_optimization_summary`
These numbers describe the *planar surrogate* only; they are not a guarantee of on-robot speed or stability.

Table 6: Optimizer-reported metrics (committed summary JSON).

Quantity	Value
Gait frequency f	1.80 Hz
Mean forward speed (steady window)	0.234 m/s
Total displacement (rollout)	1.86 m
Roll RMS	0.440 rad
Pitch RMS	~ 0 (near zero in file)
Vertical position RMS	0.086 m
Peak servo usage ratio (proxy)	0.152
Contact balance std. dev.	0.388
Dominant oscillation frequency (estimated)	3.54 Hz
Objective cost $\mathcal{J}(\boldsymbol{\phi}^{\star})$	-228.38

Reading Table 6. Mean forward speed ≈ 0.23 m/s and ≈ 1.9 m travel over the surrogate horizon indicate the optimizer found a gait that moves the reduced model forward. Roll RMS near 0.44 rad is **large**: the surrogate tolerates body oscillation in exchange for speed under the chosen weights. Pitch RMS near zero reflects the planar symmetry of the lumped model more than a claim of perfect hardware pitch stability. Peak servo usage and contact-balance spread show the solution stays inside soft actuator and support proxies, but **contact and torque models are coarse**—hardware will differ once pulses, friction, and foot slip are included.

Table 7 lists $\boldsymbol{\phi}^{\star}$ exported to `optimized_gait_params.h`. Flexion phase shifts sit at the lower bound ($-90°$): the optimizer traded stride phasing aggressively for the surrogate objective; bench tuning may move off that bound. Rest angles cluster in physically plausible ranges; frequency near 1.8 Hz matches a moderate stride rate for small hexapods in simulation-like settings.

Table 7: Optimized decision variables $\boldsymbol{\phi}^{\star}$ from L-BFGS-B (same JSON as Table 6). Angles in degrees; leg order FL, FR, ML, MR, RL, RR.

Quantity	Value
Gait frequency f	1.800 Hz
Abduction rest $\boldsymbol{\theta}_{\text{abd,rest}}$	(10, 10, 0, 0, -10, -10)
Flexion rest $\boldsymbol{\theta}_{\text{flx,rest}}$	(32.00, 32.00, 34.00, 34.00, 32.00, 32.00)
Abduction phase $\boldsymbol{\delta}_{\text{abd}}$	$\approx \mathbf{0}$ (magnitude $< 10^{-6}$ deg)
Flexion phase $\boldsymbol{\delta}_{\text{flx}}$	(-90, -90, -90, -90, -90, -90)
Swing lift amplitude	12.0 deg (`lift_swing_deg_peak`)

9.2 MuJoCo simulation (high-fidelity model)

Track II–III training uses Gymnasium with the MuJoCo asset `hexapod_training_new/assets/hexapod.xml`: checkerboard ground, twelve torque-controlled joints, and RK4 integration at 0.002 s (see configs for frame skipping). Figure 4 is an **offscreen MuJoCo render** (software `Renderer`, no interactive viewer): the model was advanced under gravity with zero actuator commands until the body settled on the plane, then a fixed camera was used. The view documents the mesh, materials, and lighting used for PPO rollouts; it is not a policy-controlled frame. Interactive inspection is available via `scripts/live_mujoco_viewer.py` with a trained checkpoint.

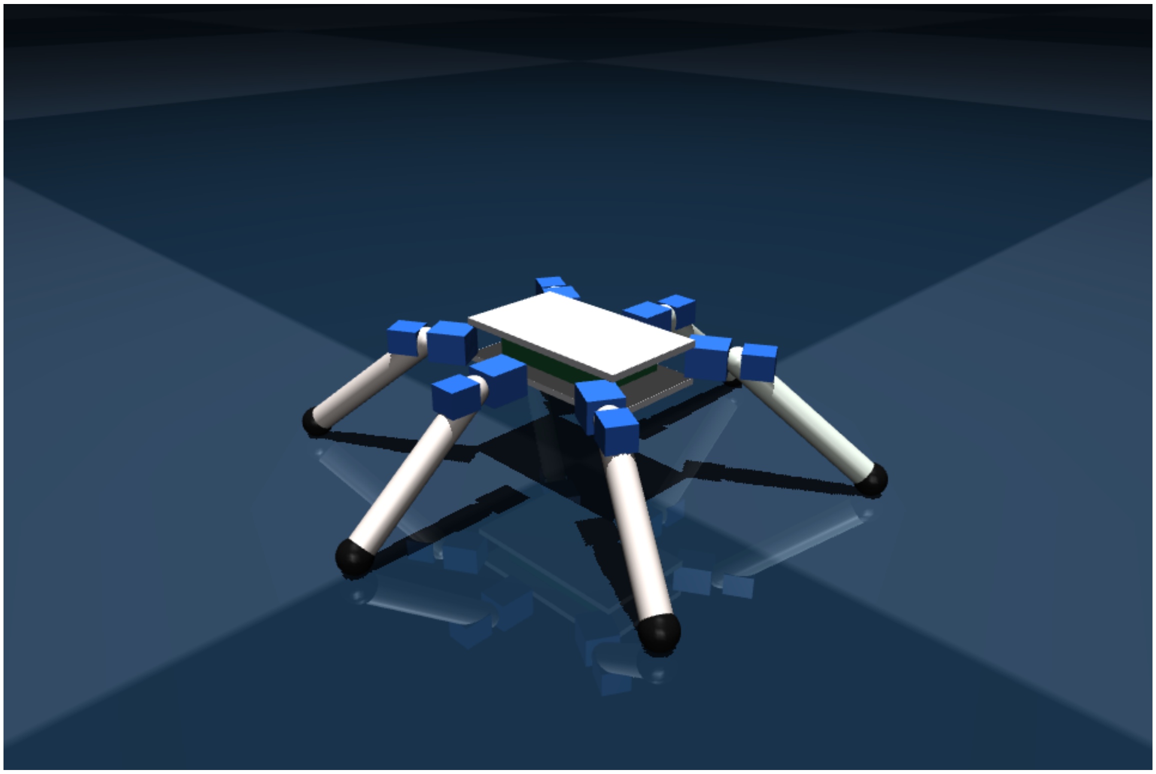


Figure 4: MuJoCo screenshot of the Walter hexapod model used for PPO training and evaluation (post-settle, zero control; offscreen render).

Figure 4 confirms the asset geometry and contact setup used for learning; it is a passive settle, not evidence of policy quality. Training and evaluation videos for specific checkpoints provide the appropriate visual readout of learned behavior.

9.3 MuJoCo PPO evaluations (released checkpoints)

Table 8 quotes short evaluation protocols documented under `e90_repo/released_checkpoints/evaluations/`. Rewards are **not** comparable across configs (different shaping and scaling); forward displacement and path length provide a common physical readout within each experiment family.

Reading Table 8. The rows are **not** comparable by raw reward (shaping and scaling differ). Forward displacement and path length are coarse physical readouts: the circle-path MLP run shows modest net forward motion and path length consistent with a curved task; the terrain run averages shorter forward progress when steps and roughness dominate. These figures support the claim that **policies were trained and evaluated in MuJoCo**, not that a single "best" locomotion score was achieved across tasks. Treat simulation displacements as **indicative**, not predictive of

Table 8: Representative PPO evaluation means (300 env steps per episode unless noted; see each REPORT.md).

Checkpoint	Task / policy	Fwd. $+x$	2D path	Notes
mlp_clean_run005_best	Circle path; MLP 256×256 PPO; training run mlp_run_005 (301k steps).	0.18 m	0.27 m	Mean reward ≈ 6800.
terrain_low_shake_run001_best	Flat, rough, and step terrain; MLP PPO.	0.33 m (3 ep. mean)	—	Flat/rough ≈ 0.47 m forward; steps harder.

hardware distance without deployment tests.

The `pd_velocity_window_run018_best` checkpoint is an sCPG + PD + velocity-window experiment with small net displacement in its released short eval—useful as an archival structured-policy datapoint, not a headline locomotion result (see that folder's `REPORT.md`).

10 Discussion

What this capstone demonstrates. The team built a **named hexapod platform** with documented mechanics, electronics, and cost; implemented an **optimization-to-firmware** path with reproducible parameters; and ran a **MuJoCo PPO training and evaluation** workflow with archived checkpoints and configs. Together, that is a credible **platform-and-methods** outcome: reusable hardware, a defensible open-loop pipeline, and simulation evidence for learned locomotion behaviors under stated rewards.

Why full adaptive locomotion is not claimed. Closing adaptive loops on hardware requires time for IMU bring-up, reflex tuning, safety testing, and power-aware deployment. The phased firmware requirements in `PHASES.md` were **not** met in the repository snapshot (Table 2); stating that once in requirements and here avoids repeating it in every technical subsection.

Technical bottlenecks. Surrogate versus MuJoCo versus hardware disagree on contact, friction, and servo dynamics; manual degree-to-pulse mapping separates optimized angles from reliable motion; open-loop gaits do not reject disturbances. RL rewards are task-specific, so **cross-comparing raw returns** across YAML files is misleading—compare displacements and behaviors within a task family only, with humility about sim-to-real.

Sim claims. MuJoCo results show that policies *can* be trained and evaluated under the project's models and rewards. They are **not** automatic predictors of field performance without transfer experiments.

Credible next steps. Harden IMU read paths and logging on the Teensy; pick one PPO checkpoint and measure a short hardware rollout with matched observation rate; then iterate reflexes or residuals before adding vision or distance-only objectives.

Advice for future teams. Lock a baseline gait and logging format before layering sensors; budget bench time after every simulation milestone; track PCB revisions and BOM multiples from week one.

11 Conclusion

This capstone **established** *Walter*: a twelve-DOF hexapod with CAD/3D-printed structure, custom PCBs, and purchased electronics totaling US$792.67 in tracked spending. The team **completed** an open-loop gait optimization loop with deterministic export to firmware, and **completed** a MuJoCo PPO training and evaluation toolchain with stored checkpoints and representative evaluations. A **bio-inspired, layered adaptive** control story informed design and simulation choices but was **not** fully realized on the MCU; that gap is documented in requirements traceability (Table 2) rather than hidden. The project provides a **strong base** for future adaptive and sim-to-real work; code and this report are maintained at `https://github.com/HowardWHSrun/WALTER`.

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A Public repository

The public project URL is `https://github.com/HowardWHSrun/WALTER`. Appendix paths below refer to the local Engineering 90 workspace layout at the time of writing; the GitHub repository is intended to subsume or mirror these components for reproducibility.

B Repository map (selected local paths)

- **Adaptive firmware and docs:** `RL Temp/adaptive_ondevice_hexapod/README.md`, `docs/DESIGN.md`, `docs/ARCHITECTURE.md`, `docs/PHASES.md`, `docs/FIRMWARE.md`
- **Firmware sketches:** `firmware/hexapod_walking_v2/`, `firmware/hexapod_openloop_sine/`, `firmware/set_zero/`, side-only tests
- **Reduced-order optimization:** `hexapod_brain_roadmap copy/ode_optimization_no_imu/`
- **MuJoCo PPO training:** `RL Temp/hexapod_training_new/` (`train.py`, `configs/`, `runs/`)
- **Released checkpoints and evals:** `RL Temp/e90_repo/released_checkpoints/`
- **Presentation notes:** `Presentation/E90_MidSemester_SpeakerScript.tex` (and PDF)
- **This folder (figures):** `walter_assembly.png`, `Body.png`, `Legs.png`, `Linkage.png`, `pcb_board.png`, `pcb_schematic.png`, `mujoco_hexapod_sim.png` (plus originals with longer filenames)